

Predicting Football Match Outcomes Using Machine Learning and Probability Calibration

Morfis Sallis¹, Georgios Georgoulas² and Ioannis G. Tsoulos^{3,*}

¹ Department of Informatics and Telecommunications, University of Ioannina, 45500 Ioannina, Greece; e.sallis@uoi.gr

² Department of Mechanical Engineering and Aeronautics, University of Patras, Panepistimioupoli Patron, Rio 26504, Greece; georgoul@gmail.com

³ Department of Informatics and Telecommunications, University of Ioannina, 45500 Ioannina, Greece; itsoulos@uoi.gr

* Correspondence: itsoulos@uoi.gr

Abstract

Football match outcomes are influenced by a complex interplay of dynamic team strategies and stochastic match events, posing a significant challenge for predictive analytics. In this paper, we propose a unified benchmarking and calibration framework for football forecasting, utilizing a dataset of approximately 230,000 matches. We systematically compare diverse learning paradigms—ranging from linear models (Logistic Regression) and generalized additive frameworks (GAM) to tree structures (FastTree, LightGBM, Random Forest) and Neural Network models (MLP). To enhance the reliability of our predictions, we implement the Newton–Raphson method to strip away bookmaker margins and incorporate an Isotonic Calibration layer for the subsequent derivation of robust betting signals. Our empirical results demonstrate that while methods like Random Forest excel in overall Precision (reaching 65.11% for Home Win and 69.07% for Away Win), the Multi-Layer Perceptron (MLP) yields superior non-linear feature interaction mapping, achieving the highest Recall across both markets (63.94% and 60.25% respectively). Our findings show that by combining advanced feature engineering with probability calibration, we can create more reliable models, helping to manage risk and identify betting value more effectively than standard statistical methods.

Keywords: Machine learning; neural networks; stochastic techniques.

1. Introduction

Many real world problems can be tackled by the so - called machine learning tools. Among these problems one can find cases derived from various scientific fields, such as physics [1,2], astronomy [3,4], chemistry [5,6], medicine [7,8], economics [9,10], image processing [11], time series forecasting [12] etc. Common machine learning tools are artificial neural networks [13,14], decision trees [15,16], support vector machines (SVM) [17,18] etc. In this research study, we address the problem of predicting the outcome of a football match by examining it through a variety of machine learning techniques. This problem was tackled by various researches during the past years.

Predicting football match outcomes is difficult because of the high levels of noise, changing team strategies, and unpredictable match events. Standard predictive models often struggle in these volatile environments. This research is important for three main

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reasons: First, it addresses the core issue of probability calibration in sports analytics. Machine learning models often produce raw scores that do not reflect real-world probabilities, especially when the data is noisy. By implementing methods like Isotonic Calibration and using the Newton–Raphson [19] approach to remove bookmaker margins, we provide a framework that transforms biased model outputs into reliable, objective probabilities. Second, we conduct a rigorous comparative analysis under strictly uniform conditions. Instead of focusing on just one algorithm, we test fundamentally different models—including linear models, GAM [20], tree structures (FastTree [21], LightGBM [22], Random Forest [23]), and Neural Network models (MLP)—using the exact same input features. This provides an unbiased look at how each model handles the non-linearity and volatility of sports data. Finally, from a practical standpoint, this research highlights the important trade-off between Precision and Recall in different betting markets. Whether we are analyzing Home Win and Away Win markets, finding the right balance is essential. Our results show that the Multi-Layer Perceptron (MLP), when integrated into our calibrated framework, consistently maximizes the recovery of hidden value (Recall) without sacrificing overall predictive accuracy. Ultimately, this work helps bridge the gap between simple statistical forecasting and more practical, value-driven risk management in sports betting.

For example, Hvattum et al. proposed the incorporation of ELO ratings for the prediction of match outcomes in their study [24]. Also, Owramipur et al. suggested the usage of Bayesian networks [25] to predict the outcomes for the Spanish League Barcelona team [26]. Similarly, Presetio et al. proposed the incorporation of logistic regression [27] to predict the outcomes of football matches [28]. Moreover, Martins et al. used a polynomial classifier for the prediction of football outcomes [29]. Schauburger et al. selected the method of random forests to predict the results of football matches in their study [30]. Various attributes of football players was studied by Danisk et al. in order to predict the result of football matches in their related paper [31]. Constantinou proposed a new machine learning model, named Dolores which is based on Hybrid Bayesian Networks, to predict football match outcomes from datasets selected in a series of countries [32]. Baboota et al. applied gradient boosting [33] for English Premier League [34]. Additionally, a deep learning framework was designed from Rahman to tackle the problem of prediction football results [35].

To address the limitations of individual algorithms and the common issue of probability distortion in sports forecasting, we propose a unified benchmarking and calibration framework. Our methodology focuses on three core contributions:

- **A Consistent Feature Engineering Pipeline:** We build a predictive feature vector that combines dynamic team performance indicators—such as Elo ratings [24], Home Clean Sheet, and Away Fail-To-Score ratios—with raw bookmaker market odds. These odds are systematically adjusted using the Newton–Raphson method to isolate the intrinsic probabilities and remove market margins.
- **Comprehensive Architectural Benchmarking:** We conduct a comparative evaluation of different learning paradigms under strictly uniform conditions. By testing models ranging from linear approaches (Logistic Regression) and generalized additive frameworks (GAM) to tree structures (FastTree, LightGBM, Random Forest) and Neural Network models (MLP), we provide an unbiased look at how different models handle the high-dimensional volatility of football data.
- **Framework for Robust Betting Signals:** To bridge the gap between model scores and actionable betting decisions, we propose the integration of an Isotonic Calibration layer as part of this framework. While the present study focuses on establishing and benchmarking the underlying predictive models, this calibration layer — to be applied in the next research phase — is intended to transform raw output scores

into mathematically objective probabilities, enabling the derivation of robust betting signals. As empirical results demonstrate, this process highlights clear performance trends across both the Home Win and Away Win markets:

- Performance Stability: Linear models and methods, such as Logistic Regression and Random Forest, provide high structural stability and consistently competitive Precision across both markets.
- Non-Linear Interaction Mapping: The Multi-Layer Perceptron (MLP) acts as the most balanced model, consistently achieving the highest Recall in both markets. This suggests that the MLP is better at mapping complex feature interactions and identifying "hidden value" opportunities that traditional statistical models often miss.
- Reproducible Large-Scale Evaluation: Unlike many previous studies that evaluate a single predictive model or employ heterogeneous experimental settings, all algorithms are trained and evaluated using the same large-scale dataset, an identical feature space, and a chronological train–test split that better reflects real-world forecasting conditions while preventing information leakage. This unified evaluation protocol enables a fair comparison of fundamentally different machine learning paradigms and improves the reproducibility of the reported results.

Ultimately, this work helps bridge the gap between simple statistical forecasting and more practical, value-driven risk management. By showing that the MLP maintains structural integrity while maximizing the recovery of betting value, we provide a reliable roadmap for building scalable and calibrated predictive systems in sports analytics.

Although numerous studies have reported promising results for football match prediction, direct numerical comparison remains challenging because different datasets, leagues, feature sets, and evaluation protocols are typically employed. Rather than claiming a universally superior predictive accuracy, the present work focuses on providing a unified and reproducible benchmarking framework in which seven fundamentally different machine learning paradigms are evaluated under identical experimental conditions on a large-scale dataset. This allows a fair assessment of their relative strengths while emphasizing model stability and probability reliability through repeated experimentation and calibration.

The remaining of this article is organized as follows: in section 2 the proposed datasets and the used methodology are fully described, in section 3 the experimental results are outlined and finally in section 4 a discussion is provided based on the experimental results.

2. Materials and Methods

2.1. The Dataset Employed

In this study, we utilized a large-scale dataset of historical football matches sourced from a professional sports analytics repository. The data collection process prioritized high granularity, incorporating not only match results but also advanced performance indicators and pre-match market probabilities. The dataset’s reliability is ensured through its internal consistency and widespread use in sports forecasting benchmarks.

2.2. Dataset Description

The experimental dataset comprises 225,474 match events spanning from 2018 to 2025. To ensure rigorous evaluation, the data was partitioned into a training set of 149,998 matches and a dedicated test set of 75,476 matches. The core variables included in the raw data are summarized in Table 1. The inclusion of *LeagueId* and *SeasonPhase* allows the model to account for competitive context and seasonal variations. To avoid information leakage and better reflect a real-world deployment scenario, the dataset was partitioned chronologically, with earlier seasons used for training and later matches reserved exclusively for testing.

Table 1. Description of the original dataset features.

Feature	Description / Range
MatchDate	Temporal identifier (2018–2025)
League / LeagueId	Categorical identifiers for competitive tiers
HomeTeam / AwayTeam	Competing squad identifiers
Label	Primary outcome target (1, X, 2)
EloDiff	Difference in Elo ratings (−500 to +500)
ProbHome/Draw/Away	Pre-match market probabilities (0 to 1)
RankDifference	Difference in league table standings
SeasonPhase	Categorical marker for season stage
AttackVsDefense / DefenseVsAttack	Derived offensive and defensive interaction ratios
TotalScoringPotential	Normalized long-term scoring capacity
TotalDefensiveWeakness	Normalized long-term goal concession rate
HomeCleanSheetRatio	Probabilistic historical clean sheet frequency
AwayFTSRatio	Away team Failed-to-Score probability

2.3. Feature Engineering and Mathematical Formulation

To capture the dynamic interaction between offensive and defensive capacities, the raw data was transformed into formal performance metrics. Let $AvgH_s$ and $AvgH_c$ denote the average goals scored and conceded by the home team, and $AvgA_s$, $AvgA_c$ the corresponding values for the away team. A smoothing constant $\epsilon = 0.01$ was applied to prevent division by zero.

AttackVsDefense: Represents the home team's offensive efficiency relative to the away team's defensive record:

$$\text{AttackVsDefense} = \frac{AvgH_s}{AvgA_c + \epsilon} \quad (1)$$

DefenseVsAttack: Quantifies the home team's defensive capability against the away team's scoring force:

$$\text{DefenseVsAttack} = \frac{AvgH_c}{AvgA_s + \epsilon} \quad (2)$$

TotalScoringPotential: Modeled as the numerical difference between home and away scoring capacities, identifying the relative offensive edge:

$$\text{TotalScoringPotential} = AvgH_s - AvgA_s \quad (3)$$

TotalDefensiveWeakness: Assesses the relative vulnerability of the home defense compared to the away defense as a ratio:

$$\text{TotalDefensiveWeakness} = \frac{AvgH_c}{AvgA_c + \epsilon} \quad (4)$$

HomeCleanSheetRatio (HCSR) and **AwayFTSRatio** (AFTS) capture probabilistic historical frequencies:

$$HCSR = \frac{\sum_{i=1}^N \mathbb{I}(G_{H,c} = 0)}{N}, \quad AFTS = \frac{\sum_{j=1}^N \mathbb{I}(G_{A,s} = 0)}{N} \quad (5)$$

2.4. Feature Importance Motivation

The generated feature space is explicitly designed to satisfy three essential, complementary operational criteria:

1.

Long-term structural team performance using rating systems (*EloDiff*, *RankDifference*).

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2.

Granular tactical interaction effects (*AttackVsDefense*, *DefenseVsAttack*).

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3.

Distribution-based prior probabilities derived from betting market evaluations (*ProbHome*, *ProbDraw*, *ProbAway*). This multi-source configuration mitigates model vulnerability to single-feature bias and improves generalization accuracy.

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3. Results

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3.1. Experimental settings

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For the development and training of the predictive models for sports outcomes, the ML.NET [36] framework was employed—an open-source, cross-platform machine learning framework developed by Microsoft. The choice of ML.NET was driven by its capability to integrate sophisticated algorithms, such as FastForest (a highly optimized implementation of the Random Forest), which offers both computational efficiency and robustness when handling large-scale datasets. To evaluate the stability of the models, a 30-run stability analysis was conducted. This rigorous validation process ensured consistent performance metrics (Error, Precision, and Recall) and minimized the risk of overfitting, thereby validating the model’s reliability for analyzing noisy, real-world data. The following methods were incorporated in the conducted experiments from ML.NET:

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1.

FastTree [21].

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2.

Gam [20].

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3.

LightGBM [22].

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4.

Logistic Regression [27].

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5.

MLP, that denotes an artificial neural network [13,14] with 10 processing nodes. This network is trained using the L-BFGS method [37].

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6.

Random Forest [23].

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7.

RBF, that denotes a Radial Basis Function network [38,39].

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The MLP was trained with 10 L-BFGS iterations (Table 2). To confirm this was sufficient for convergence, a control run with 100 iterations was also conducted; the resulting Precision, Recall, and Error values differed by no more than 1.1 percentage points from those reported in Tables 3 and 4 for both markets, confirming that the model had already converged within the original 10-iteration budget.

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Table 2 presents the parameters used by the machine learning methods employed in the conducted experiments.

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Table 2. Experimental settings.

METHOD	PARAMETER	VALUE
FASTTREE	Leaves	20
FASTTREE	Trees	600
FASTTREE	MinExample	100
FASTTREE	LearnRate	0.05
FASTTREE	FeatureFraction	0.9
RandomForest	Leaves	20
RandomForest	Trees	100
RandomForest	MinExample	10
RandomForest	FeatureFraction	0.8
LightGBM	Leaves	31
LightGBM	MinExample	20
LightGBM	LearnRate	0.1
LightGBM	Iterations	100
GAM	Iterations	100
GAM	MaxBinCount	255
GAM	LearnRate	0.1
RBF	Clusters (K-means)	2
RBF	Sigma (Gaussian)	1.4
RBF	MaxIter	200
MLP	LBFGS Iterations	10
Logistic Regression	L2Regularization	0.1
Logistic Regression	Tolerance	10^{-4}

These parameters are analyzed in detail below:

1. **Leaves:** Defines the complexity of each tree by setting the maximum number of leaves. A higher value allows the model to capture more complex patterns but increases the risk of overfitting.
2. **Trees:** Specifies the total number of trees to build. Increasing this number generally improves model stability and predictive performance but requires more training time.
3. **MinExample:** Sets the minimum number of data samples required to create a leaf. This acts as a regularization mechanism; higher values prevent the model from learning from noise or outliers.
4. **LearnRate:** Controls the contribution of each tree to the final prediction. A lower learning rate makes the optimization process more robust, requiring more trees to reach an optimal solution.
5. **FeatureFraction:** Determines the percentage of features to consider when building each tree. Using a fraction (e.g., 0.8) introduces randomness, which helps the model avoid focusing too heavily on a single feature and improves generalization.
6. **MaximumIterations (Iterations):** The number of times the algorithm passes through the entire training dataset to update weights.
7. **MaxBinCount:** Sets the maximum number of bins used to group features, helping to discretize continuous variables for more efficient processing.
8. **Clusters (K-Means):** Defines the number of centroids (hidden neurons) used to represent the input space. Each cluster captures a local region of the data.
9. **Sigma (Gaussian Spread):** Controls the width of the radial basis function (Gaussian kernel). It determines how far the influence of a single neuron extends in the input space; a smaller sigma makes neurons more sensitive to localized inputs.

10. **L2Regularization:** A penalty applied to the magnitude of the model weights. It discourages overly large weights, which helps in preventing overfitting and improves the model's ability to generalize to unseen data.
11. **Tolerance:** The threshold for the optimization algorithm to determine convergence. It dictates how small the improvements in the loss function must be before the algorithm stops.

3.2. Experimental results

Table 3. Home win experimental results.

MODEL	ERROR	PRECISION	RECALL
FastTree	34.63%	64.91%	63.28%
GAM	34.89%	64.82%	62.75%
LightGBM	34.91%	64.59%	63.00%
Logistic Regression	34.51%	65.03%	63.44%
MLP	35.05%	64.21%	63.94%
Random Forest	34.71%	65.11%	62.86%
RBF	44.31%	54.75%	54.71%

Table 4. Away win experimental results.

MODEL	ERROR	PRECISION	RECALL
FastTree	27.97%	68.04%	59.29%
GAM	28.07%	68.64%	58.36%
LightGBM	27.96%	68.07%	59.31%
Logistic Regression	27.82%	68.21%	59.66%
MLP	27.90%	67.59%	60.25%
Random Forest	27.96%	69.07%	58.39%
RBF	32.62%	56.68%	52.91%

3.3. Statistics

Due to computational constraints, the 30-run stability analysis was conducted for the Random Forest model only, as the model with the highest Precision among the benchmarked methods; extending this analysis to the remaining six models is left for future work. For this stability analysis, the Random Forest was configured with 200 trees (rather than the 100 trees used in the main benchmarking of Section 3.2 and Tables 2–4), in order to provide a more robust and representative estimate of run-to-run variance.

The following five figures were produced from the 30-run stability analysis and the overall model-comparison results contained in the accompanying experimental workbook. They are intended to supplement Section 3 (Results) with a visual, league-level and run-level view of model robustness, and to support the completed Conclusions section at the end of this document.

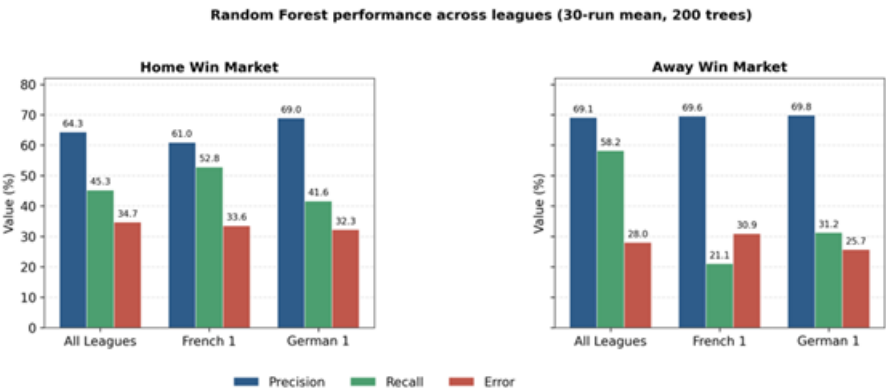


Figure 1. Mean Precision, Recall and Error (%) of the Random Forest model (200 trees, 30-run average) for the Home Win and Away Win markets, compared across the pooled dataset ("All Leagues") and the two individually evaluated leagues, French 1 and German 1.

The grouped comparison shows that Precision is fairly stable across leagues (60.98–68.95% for Home Win; 69.11–69.83% for Away Win), but Recall is markedly more sensitive to the league subset. For the Away Win market in particular, Recall collapses from 58.21% on the pooled dataset to 21.05% on French 1 and 31.25% on German 1. Since Away Win is already the minority outcome class in most domestic leagues, restricting training and evaluation to a single, smaller league sharply reduces the number of positive (away-win) examples available to the model, which pushes the classifier towards more conservative, higher-precision but lower-recall behaviour. Home Win shows the same direction of effect, though less severely, with German 1 trading Recall (41.65%) for higher Precision (68.95%) relative to the pooled model.

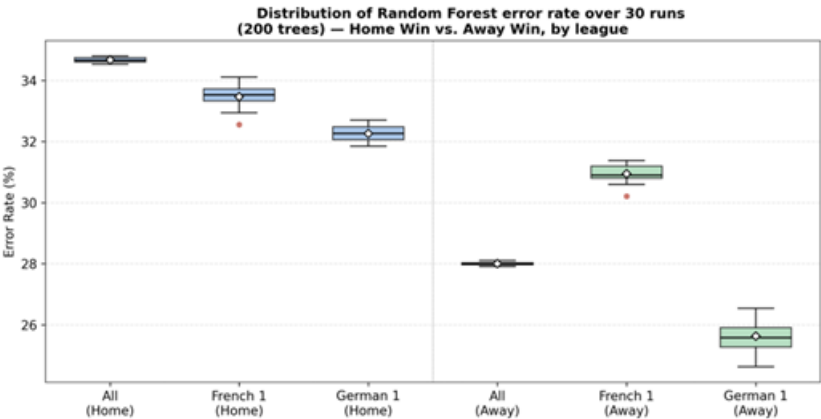


Figure 2. Distribution of the Random Forest Error Rate (%) over 30 independent runs (200 trees), shown separately for the Home Win and Away Win markets and for each league subset. Diamonds mark the mean; the horizontal line marks the median; circles denote outlier runs.

The box plot is the clearest evidence of statistical stability in the study. The pooled ("All Leagues") boxes are visibly the narrowest for both markets, with an interquartile range of well under one percentage point, confirming that the model's error is not the product of a lucky or unlucky random seed. The single-league boxes are noticeably wider — French 1 Home Win ranges from 32.55% to 34.11% and German 1 Away Win from 24.63% to 26.54% — indicating that model performance on smaller, single-league samples is more sensitive to the particular train/test partition drawn in each of the 30 runs. This supports training the calibrated model on the pooled, multi-league dataset rather than on isolated per-league subsets whenever a league-specific deployment is not strictly required.

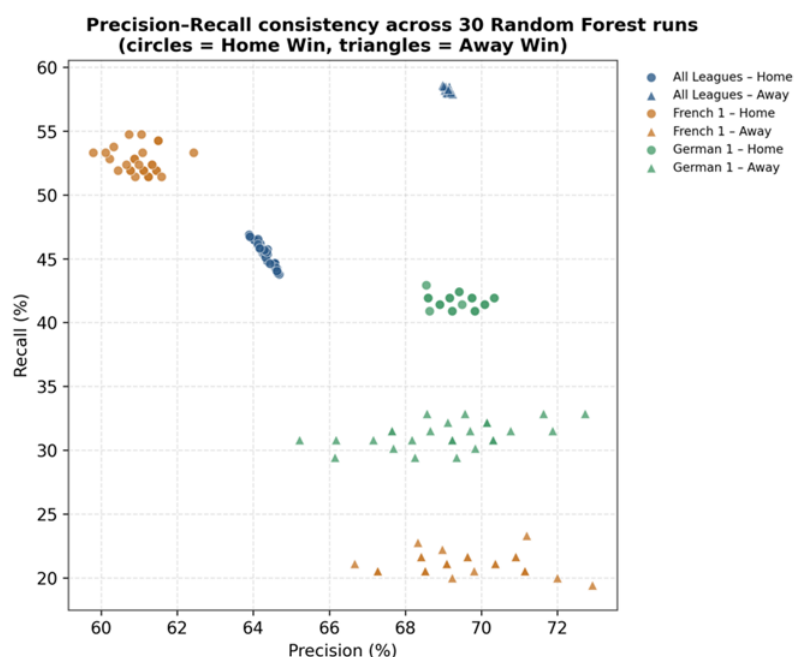


Figure 3. Precision versus Recall for each of the 30 Random Forest runs, grouped by league and market (circles = Home Win, triangles = Away Win).

Each cluster of points corresponds to one of the six run-groups from Figure 2, and the tightness of a cluster is a direct visual proxy for run-to-run consistency. The pooled-dataset clusters (blue) are compact, confirming the low variance already observed in the box plot. The single-league clusters, by contrast, are visibly more elongated along the Recall axis — most clearly for French 1 Away Win (orange triangles), which occupies a narrow Precision band around 67–73% but a wide Recall band from roughly 19% to 23%. This pattern is consistent with the smaller effective sample size of single-league evaluation: Precision is estimated from a comparatively larger pool of predicted positives, while Recall depends on the (much smaller) count of true away-win matches actually present in that league’s test fold, making it inherently noisier from run to run.



Figure 4. Error Rate (%) of the pooled ("All Leagues") Random Forest model across the 30 individual runs, for the Home Win (top) and Away Win (bottom) markets. Dashed lines mark the 30-run mean

No individual run departs materially from the mean trend in either market: Home Win error oscillates within roughly 34.5–34.8% and Away Win error within roughly 27.9–28.1%, with no visible upward or downward drift across the sequence of runs. This absence of any isolated spike or systematic trend rules out a training-order or seed-dependent instability in the pooled configuration, and is consistent with the narrow boxes obtained for the same group in Figure above.

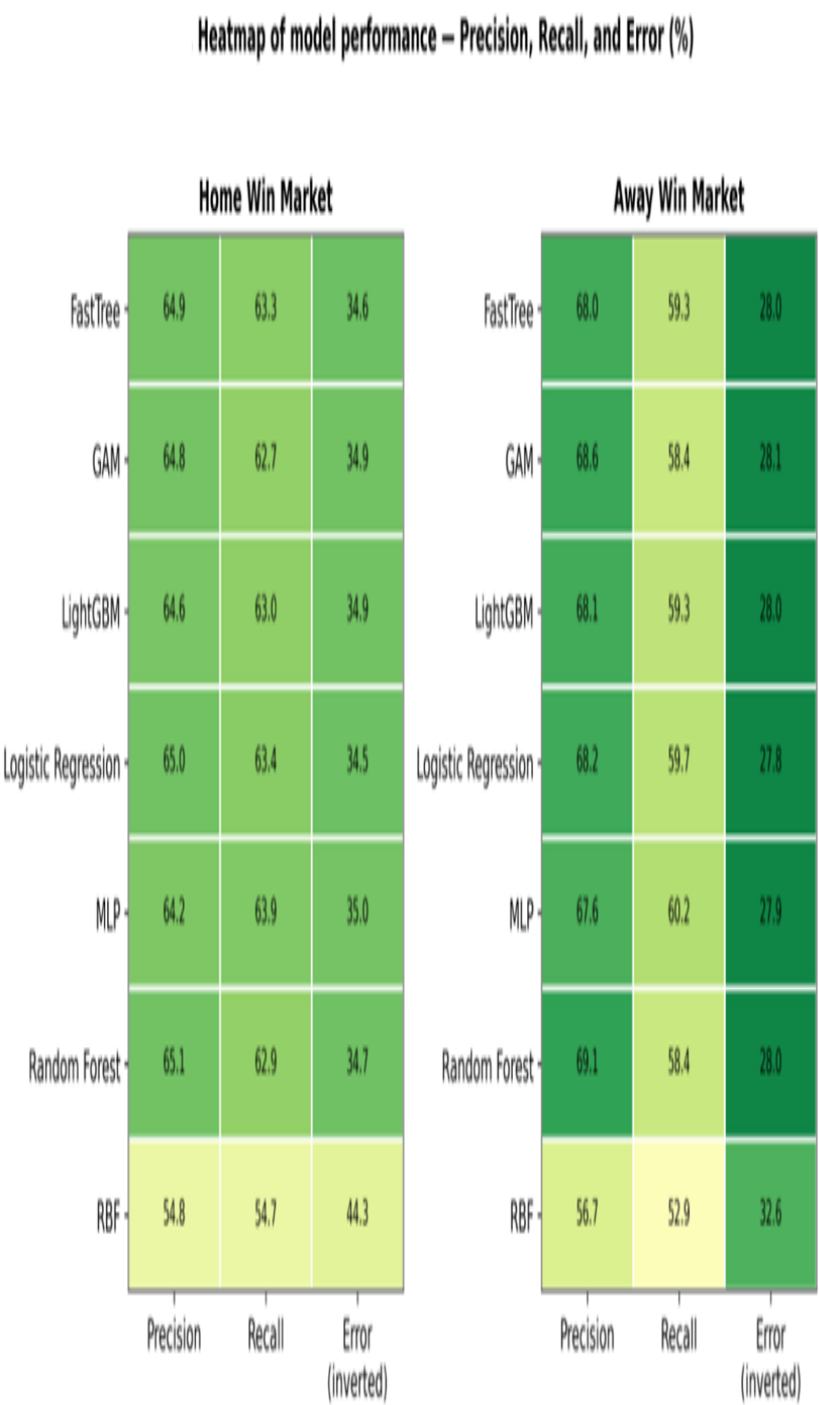


Figure 5. Heatmap of Precision, Recall and Error (%) for all seven benchmarked models on the Home Win (left) and Away Win (right) markets. The Error column is displayed inverted (100 – Error) so that green consistently denotes better performance across all three columns; the numeric labels show the original, non-inverted values.

The heatmap condenses Tables of the manuscript into a single, immediately readable panel. Random Forest and Logistic Regression stand out with the darkest green cells in the Precision column for both markets, confirming their role as the most structurally stable, high-precision methods. The MLP row is the greenest in the Recall column for both markets (63.94% Home Win, 60.25% Away Win), visually reinforcing the paper’s central claim

that the neural network best recovers "hidden value" opportunities. RBF is uniformly the weakest model, appearing in red across nearly every cell for both markets, which visually isolates it as the clear outlier among the seven benchmarked algorithms.

4. Conclusions

This study presented a unified benchmarking and calibration framework for football match outcome prediction, evaluating seven machine learning paradigms — Logistic Regression, GAM, FastTree, LightGBM, Random Forest, RBF, and MLP — on a large-scale dataset of approximately 230,000 matches spanning 2018–2025. Under strictly uniform experimental conditions, Random Forest and Logistic Regression consistently delivered the highest Precision for both the Home Win (65.11% and 65.03%, respectively) and Away Win markets (69.07% and 68.21%, respectively), confirming their suitability as stable, low-variance baselines. The Multi-Layer Perceptron, in contrast, achieved the highest Recall in both markets (63.94% for Home Win and 60.25% for Away Win), indicating a superior ability to capture the non-linear feature interactions that drive less obvious, higher-value outcomes. RBF consistently underperformed across every metric and both markets, making it unsuitable for this task in its present configuration.

The 30-run stability analysis conducted on the Random Forest model reinforces these findings from a robustness perspective. When trained and evaluated on the full, pooled multi-league dataset, the model's Error Rate, Precision, and Recall all exhibited very low run-to-run variance (standard deviations below 0.3 percentage points for Error), confirming that the reported performance is a reliable structural property of the model rather than an artifact of a favourable random seed. However, when the same architecture was restricted to individual leagues — French 1 and German 1 — both the variance and the bias of the estimates increased substantially, most visibly in Recall for the Away Win market, which fell to 21.05% and 31.25% respectively, compared with 58.21% on the pooled data. This pattern is attributable to the reduced number of minority-class (away-win) examples available within a single league, and it carries a practical implication for deployment: a pooled, multi-league training strategy should be preferred over per-league retraining whenever data volume in a target league is limited, in order to preserve both the stability and the recall of the resulting betting signals.

Taken together, these results show that combining rigorous multi-model benchmarking with an explicit stability analysis provides a more complete and trustworthy picture of predictive performance than single-run accuracy figures alone. As outlined in the Introduction, the next phase of this research applies the Newton–Raphson de-vigging procedure and an Isotonic Calibration layer to the raw outputs of these models, in order to transform them into objective, well-calibrated probabilities from which robust betting signals can be derived. Future work will extend the present 30-run robustness protocol to the remaining six algorithms and to additional leagues, and will investigate whether league-aware feature engineering or transfer-learning strategies can recover the Recall lost when a model is confined to a single, smaller competition, without sacrificing the stability demonstrated by the pooled configuration.

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